

Minor Exam – 1 (Section A)

Course Code: EC150

Total Marks: 25

Course Name: Basics of Electronics Engineering

Time Duration: 1 Hour

Answer ALL the Questions

1. Describe in your own words the conditions established by forward- and reverse-bias conditions on a p-n junction diode and how the resulting current is affected. (4M)
 2. Determine the diode current at 20°C for a silicon diode with $I_s = 50\text{ nA}$ and an applied forward bias of 0.6 V . (3M)
 3. Design and draw a suitable circuit to glow the LED of its turn on voltage is 2.2 V and the Input source voltage is 5 V and current is 0.5 mA . (5M)
 4. A full-wave bridge rectifier with a 120 V_{rms} sinusoidal input has a load resistor of $1\text{ k}\Omega$.
 - (a) If silicon diodes are employed, what is the dc voltage available at the load? (2M)
 - (b) Determine the required PIV rating of each diode. (2M)
 - (c) Find the maximum current through each diode during conduction. (2M)
 - (d) What is the required power rating of each diode? (2M)
 5. With neat circuit diagram explain how the BJT transistor can be acted as a switch. (5M)
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25/5/24

Minor Exam – 2 (Section C & D)

Course Code: EC150

Total Marks: 25

Course Name: Basics of Electronics Engineering

Time Duration: 1 Hour

Answer ALL the Questions

1. Find the octal equivalent of $(173.55)_{10}$. (3M)
2. Obtain the binary equivalent of $(1FE.F6)_{16}$ and the hex equivalent of $(1111001110.011111001)_2$. (5M)
3. The carrier power is 10 kW. Find the maximum possible power in the AM signal if it is a conventional AM signal. (3M)
4. The output of the AM modulator is given as $s(t) = 20\cos 2000\pi t + 5\cos 1800\pi t + 5\cos 2200\pi t$. The amplitude sensitivity $k_a = 0.25$ V. Calculate modulation index, find message signal, and carrier signal. (5M)
5. Draw a voltage follower circuit using an OP-AMP and explain why it is called a voltage follower. (3M)
6. Design a circuit using a single OP-AMP to realize the mathematical expression $v_o(t) = 5v_1(t) - 3v_2(t)$. (6M)



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech./M.Tech./Ph.D.

Mid Semester Examinations, October-2025

Course Name: Basics of Electronics Engineering

Date: 13-10-2025

Duration: 1.5 Hours

Course Code: EC150

Time: 02.00 – 03.30 PM

Max. Marks: 50

ANSWER ALL QUESTIONS

1. Determine the output waveform for the circuit shown in Figure 1. Assume the diode is made up of silicon. (4M)

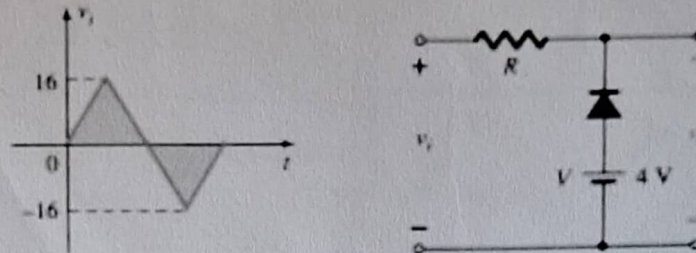


Figure 1. Input and circuit for Problem 1.

2. With neat circuit diagram explain the working principle of Full wave rectifier using PN junction diodes. Also draw the input and output waveforms. (6M)
3. Determine V_o and I for the circuits shown in Figure 2(a) and (b). (6M)

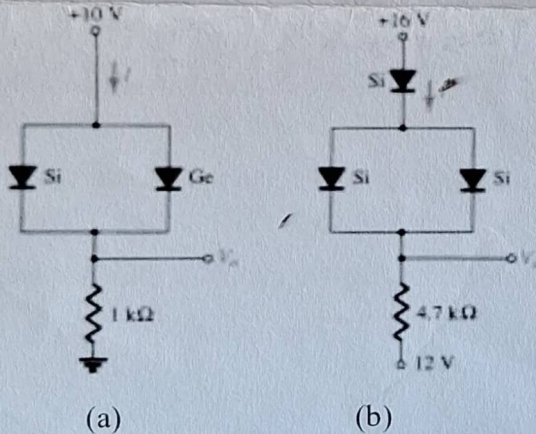


Figure 2. Circuit for Problem 3

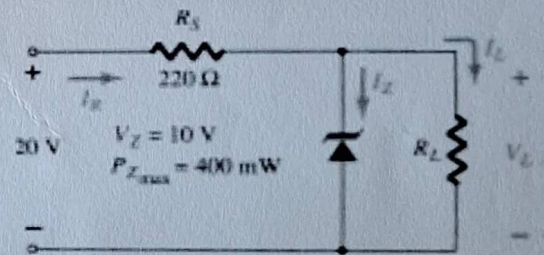


Figure 3. Circuit for Problem 4

4. (a) Determine V_L , I_L , I_Z , and I_R for the network shown in Figure 3. if $R_L = 180\Omega$. (4M)
(b) Repeat part (a) if $R_L = 470\Omega$. (4M)
(c) Determine the value of R_L that will establish maximum power conditions for the Zener diode. (2M)
(d) Determine the minimum value of R_L to ensure that the Zener diode is in the "on" state. (2M)
5. List out the major difference between a bipolar and a unipolar device? (4M)
6. (a) Using the characteristics of Common Emitter configuration shown in Figure 4, determine β_{dc} at $I_B = 80\mu A$ and $V_{CE} = 5V$. (2M)
(b) Repeat part (a) at $I_B = 5\mu A$ and $V_{CE} = 15V$. (2M)
(c) Repeat part (a) at $I_B = 30\mu A$ and $V_{CE} = 10V$. (2M)

(d) Reviewing the results of parts (a) through (c), does the value of β_{dc} change from point to point on the characteristics? Where were the higher values found? Can you develop any general conclusions about the value of β_{dc} on a set of characteristics such as those provided in Figure 4? (6M)

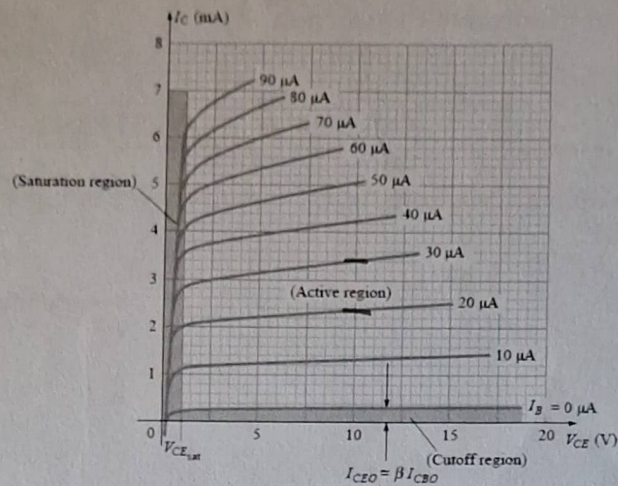


Figure 4. Output Characteristics of CE configuration for Problem 6

7. (a) Draw the basic construction of a n -channel JFET. (3M)
- (b) Apply the proper biasing between drain and source and sketch the depletion region for $V_{GS} = 0V$. (3M)

(6)

All the Best

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National Institute of Technology Goa

Programme Name: B.Tech.

Mid-semester Examination, March 2026

Course Name: **Basics of Electronics Engineering**

Date: 07/03/2026

Duration: 1Hr 30Min.

Course Code: **EC150**

Time: 10.00AM - 11:30AM

Max. Marks: 50

Attempt all the questions.

1. Draw the simplest half-wave rectifier circuit using a diode. For a sinusoidal input signal, obtain the time-average (dc) value of the output waveform of the half-wave rectifier. Consider both scenarios: (i) an ideal diode and (ii) a diode with forward resistance. If the input signal frequency is 1.8 GHz and the peak amplitude ranges from 2 μ V to 10 mV, find the range of the output average for the ideal diode case. [10]
2. Draw the structure of *npn*- type BJT and its symbol. Explain the transistor action in active mode of operation with all the related equations governing the operation and performance parameters. Also, draw the small-signal model for the same with base width modulation. [8]
3. Compare half-wave and full-wave rectifiers. Design a full-wave rectifier to deliver an average power of 2W to a cellphone with a voltage of 3.6 and ripple of 0.2. [8]
4. Calculate small-signal parameters of a MOSFET which is biased at a drain current of 0.5mA. Assume $\mu_n C_{ox} = 100 \mu A/V^2$, $V_{TH} = 0.5V$, $\lambda = 0.1V^{-1}$, and $W/L = 10$. [8]
5. An AM signal with carrier frequency $f_c = 1$ MHz is modulated by an audio signal whose highest frequency is $f_m = 5$ kHz. The signal is to be demodulated using an envelope detector consisting of a diode followed by a parallel *R-C* circuit. [8]
 - (a) Draw the circuit and state the conditions on time constant *RC* for proper envelope detection.
 - (b) If the load resistor is $R = 10$ k Ω , find a suitable value of the capacitor *C*.
6. A narrowband FM signal is given by [8]

$$s(t) = 10 \cos(2\pi \cdot 10^6 t) - 3 \sin(2\pi \cdot 10^6 t) \sin(2\pi \cdot 2 \times 10^3 t),$$

determine: (a) Carrier frequency f_c , (b) Expression of message $m(t)$, (c) Message frequency f_m , and (d) Modulation index β_F .

ALL THE BEST



Roll No 25CSE1049

NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

End Semester Examinations, December-2025

Course Name: Basics of Electronics Engineering

Course Code: EC150

Date: 15-12-2025

Time: 10.00 AM - 01.00 PM

Duration: 3 Hours

Max. Marks: 100

ANSWER ALL QUESTIONS

1. Determine V_o , I_1 , I_{D1} , and I_{D2} for the parallel diode (Si) configuration of Fig. 1. (5M)

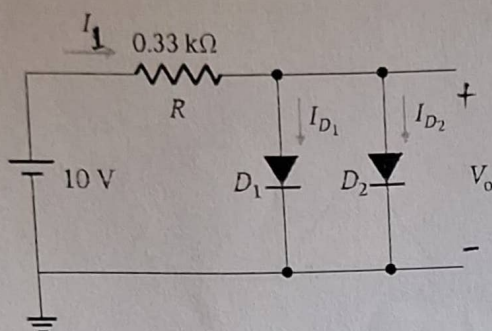


Fig. 1 Diode circuit (Problem 1)

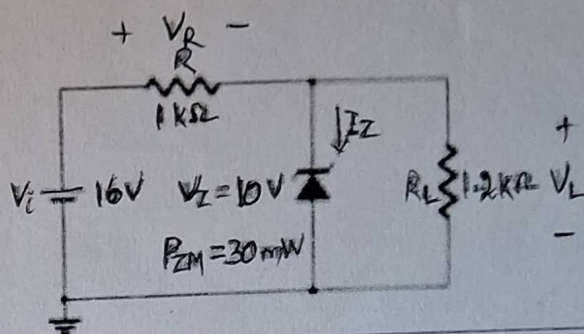


Fig. 2 Zener diode network (Problem 2)

2. (a) For the Zener diode network of Fig. 2, determine V_L , V_R , I_Z , and P_Z . (5M)
(b) Repeat part (a) with $R_L = 3k\Omega$. (5M)
3. Given $I_{DSS} = 9mA$ and $V_P = -3.5V$ for JFET, determine I_D when: (a) $V_{GS} = 0V$, (b) $V_{GS} = -2V$, and (c) $V_{GS} = -3.5V$. (6M)
4. Explain the basic construction, operation and characteristics p-channel enhancement type MOSFET with neat diagram. (10M)
5. Draw the truth table of the logic circuit shown in Fig. 3. (6M)

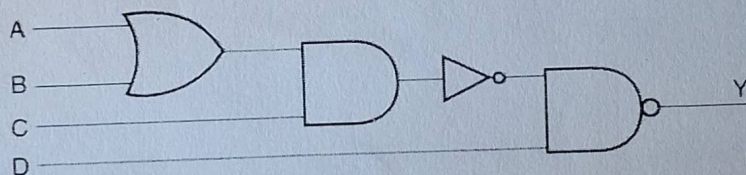


Fig. 3 Logic circuit (Problem 5)

6. Why are NAND and NOR gates called universal gates? Justify your answer with the examples. (5M)
7. Simplify the Boolean expression given by $F(A, B, C, D) = \Sigma(0, 2, 4, 6, 8, 10, 12, 14) + \Sigma_d(3, 7, 11, 15)$ and draw its logic circuit diagram. (10M)
8. Design the digital circuits for a half subtractor and a full subtractor. (10M)
9. List out the characteristics of Ideal and Practical OPAMP. (6M)

10. Design a suitable amplifier circuit for the output expression is given by $V_o = -(0.2V_1 + 0.8V_2 + 1.2V_3)$ using a single OPAMP. Assume $R_F = 10k\Omega$. Draw the circuit neatly. (6M)

11. For the circuit shown in Fig. 4, what is the condition of each of the LEDs for (i) $v_i = 0.5V$, (ii) $v_i = 3V$. (6M)

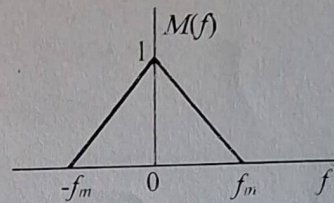
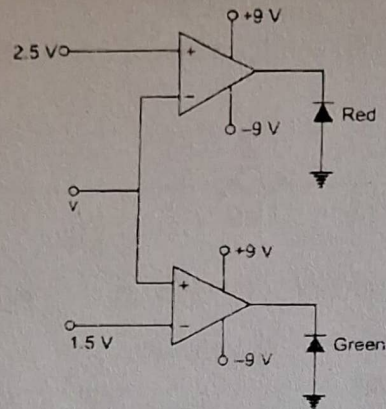


Fig. 4 OPAMP circuit (Problem 11)

Fig. 5 Spectrum of $m(t)$ (Problem 13)

12. What is modulation? Explain in detail the different modulation schemes used in communication systems. (10M)
13. Consider a message signal $m(t)$ with the spectrum shown in Fig. 5, where $f_m = 1\text{ kHz}$. This signal is applied to a product modulator, together with the carrier wave $A_c \cos(2\pi f_c t)$, producing DSB-SC modulated wave $s(t)$. Determine the spectrum of the modulated signal when: (i) $f_c = 1.5\text{ kHz}$, and (ii) $f_c = 0.5\text{ kHz}$. (10M)

All the Best